Microsoft Endpoint Detection and Response

Introduction and Fundamentals

- Introduction to Cyber Security and EDR
- Overview of Microsoft Defender for Endpoint (MDE)
- Key Concepts:
  - Threat and Vulnerability Management,
  - Attack Surface Reduction (ASR)

Labs

- Basic Configuration and Policy Settings

Microsoft Endpoint Detection and Response (EDR)

- EDR is a cybersecurity technology that **continuously monitors endpoints for evidence of threats and performs automatic actions** to help mitigate them.
  - Endpoints—the many physical devices connected to a network, such as mobile phones, desktops, laptops, virtual machines, and Internet of Things (IoT) technology—give malicious actors multiple points of entry for an attack on an organization.
- EDR solutions **help security analysts detect and remediate threats** on endpoints before they can spread throughout your network.
- EDR security solutions log behaviors on endpoints around the clock.
- They continuously analyze this data to reveal suspicious activity that could indicate threats such as ransomware.

What is Microsoft Defender for Endpoint (MDE)?

Defender for Endpoint is an enterprise endpoint security **platform** that uses the following combination of technology built into Windows and Microsoft's robust cloud service:

1. **Endpoint behavioral sensors**: Embedded in Windows, these sensors collect and process behavioral signals from the operating system and send sensor data to your private, isolated, cloud instance of Defender for Endpoint.

What is Defender for Endpoint?

2. Cloud security analytics: Using big data, device learning, and unique Microsoft optics across the Windows ecosystem, enterprise cloud products (such as Microsoft 365), and online assets, behavioral signals are translated into insights, detections, and recommended responses to advanced threats.

3. Threat intelligence: Generated by Microsoft hunters and security teams, and augmented by threat intelligence provided by partners, threat intelligence enables Defender for Endpoint to identify attacker tools, techniques, and procedures, and generate alerts when they're observed in collected sensor data.

Microsoft Defender for Endpoint

- Microsoft Defender for Endpoint is an enterprise endpoint security platform designed to help enterprises **prevent, detect, investigate, and respond** to advanced threats on their endpoints.
- Endpoints include laptops, phones, tablets, PCs, access points, routers, and firewalls.
- Defender for Endpoint is part of Microsoft Defender XDR and can be integrated with other Microsoft solutions, including:
  - Intune
  - Microsoft Defender for Cloud
  - Microsoft Defender for Cloud Apps
  - Microsoft Defender for Identity
  - Microsoft Defender for Office
  - Microsoft Defender Vulnerability Management
  - Microsoft Sentinel

Defender for Endpoint supports:

- Windows
- MacOS
- Linux
- Android
- iOS

Pre-requisites – Defender for Endpoint

- To onboard servers to Defender for Endpoint, licenses are required.
- Updated web browser for accessing Defender for endpoint and Microsoft Defender XDR.
- Supported windows versions:
  - Windows 10 and 11 **Enterprise, IoT Enterprise, Education, Pro, Pro Education**.
  - Windows Server 2012 and later (2012R2, 2016, 2019, 2022 and 2025).
    - Including editions like:
      - Standard, Enterprise, Essential, Foundation.
  - Windows 365 Cloud PCs and supported Azure (Windows) Virtual Desktop.

Pre-requisites – Defender for Endpoint

- Other operating systems supported by Defender for Endpoint
  - Mac (client devices), Linux
  - Windows Subsystem for Linux (WSL)
  - Android, iOS
- Hardware requirements:
  - Cores: 2 minimum, 4 preferred
  - Memory: 1GB minimum, 4 GB preferred
- Internet connectivity on devices is required either directly or through a proxy.
- The Defender for Endpoint agent depends on Microsoft Defender Antivirus to scan files and provide information about them.

Licensing

Defender for Endpoint is available in three plans.

- **Microsoft Defender for Business**
  - An endpoint security solution designed for small and medium-sized businesses.
- **Defender for Endpoint Plan 1 (P1)**
  - Provides core endpoint protection capabilities.
- **Defender for Endpoint Plan 2 (P2)**
  - A comprehensive endpoint protection solution that includes advance capabilities.

Set up your trial

1. Confirm your license state.
2. Set up role-based access control and grant permissions to your security team.
3. Visit the Microsoft Defender portal.
4. Onboard endpoints using any of the supported management tools.
5. Configure capabilities.
6. Visit the Microsoft Defender portal.

Portals to be used:

- **Microsoft Defender for Endpoint →** https://security.microsoft.com
- **Microsoft Intune Portal →** https://endpoint.microsoft.com
- **Microsoft 365 Admin Portal →** https://admin.microsoft.com
- **Microsoft 365 Portal →** https://m365.cloud.microsoft
- **Microsoft Azure →** https://portal.azure.com
- Prepare to deploy Microsoft Defender for Endpoint deployment

Core Objectives of Microsoft EDR

- Microsoft EDR is designed to:
  - Detect sophisticated and fileless attacks
  - Reduce attacker dwell time
  - Provide full attack-chain visibility
  - Enable rapid containment and remediation
  - Integrate endpoint security into a broader XDR strategy

Microsoft Defender
for Endpoint

How does EDR work?

While EDR technology may vary with each vendor, they work in broadly the same way. An EDR solution:

- **Continuously monitors endpoints.**
  - The EDR installs a software agent on newly boarded devices to ensure the whole digital ecosystem is visible to security teams. This software agent continuously logs relevant activity on each managed device.
- **Aggregates telemetry data**
  - The data ingested from each device is sent back from the agent to the EDR solution, which can be in the cloud or on-premises.
  - Event logs, authentication attempts, application use, and other information are made visible to security teams in real time.

How does EDR work?

- **Analyzes and correlates data**
  - The EDR solution uncovers IOCs that would otherwise be easy to miss.
  - EDRs typically use AI and machine learning to apply behavioral analytics based on global threat intelligence to help your team fend off advanced tactics being used against your organization.
- **Surfaces suspected threats and takes automatic remediation actions**
  - EDR solution flags a potential attack and sends an actionable alert to your security team so they can respond quickly.
  - Depending on the trigger, the EDR system may also isolate an endpoint or otherwise contain the threat to prevent it from spreading while the incident is being investigated.
- **Stores data for future use**
  - EDR technology keeps a forensic record of past events to inform future investigations.
  - Security analysts can use this to consolidate events or to get the big picture about a prolonged or previously undetected attack.

Integration with Microsoft Security Ecosystem

- Microsoft EDR integrates tightly with:
  - Microsoft Defender Antivirus
  - Microsoft Defender for Identity
  - Microsoft Defender for Office 365
  - Microsoft Sentinel (SIEM)
  - Microsoft XDR
- This enables cross-domain attack correlation.

Threats Detected by Microsoft EDR

- Ransomware and malware
- Credential theft
- Lateral movement
- Command-and-Control (C2)
- Privilege escalation
- Insider threats
- Advanced Persistent Threats (APT)

Licensing

- Available via:
  - Microsoft Defender for Endpoint P2
  - Microsoft 365 E5
  - Windows E5
- Feature availability depends on license tier.

Overview of Microsoft Defender for Endpoint (MDE)

- Microsoft Defender for Endpoint (MDE) is a cloud-native, enterprise-grade Endpoint Detection and Response (EDR) platform designed to prevent, detect, investigate, and respond to advanced threats across endpoints.
- It forms a core component of Microsoft XDR, integrating endpoint security with identity, email, cloud, and application protection.
- Supported Platforms
  - Windows 10 / 11
  - Windows Server
  - macOS
  - Linux
  - Android
  - iOS

Microsoft Defender for Endpoint is designed to:

- Protect endpoints from advanced cyber threats
- Provide deep, continuous endpoint visibility
- Detect fileless and zero-day attacks
- Reduce attacker dwell time through rapid response
- Enable proactive threat hunting

Key Components

1. Endpoint Sensor

- Built into Windows 10/11 and Windows Server
- Collects behavioral telemetry:
    - Process execution
    - File and registry changes
    - Network connections
    - PowerShell and script activity

2. Cloud-Based Analytics

- Machine learning and behavioral analysis
- Microsoft global threat intelligence
- Correlates endpoint events into incidents

3. Microsoft Defender Portal

- Centralized management console
- Incident investigation and response
- Advanced hunting and reporting

Key Concepts

- Threat and Vulnerability Management (TVM) & Attack Surface Reduction (ASR)
  - Microsoft Defender for Endpoint (MDE) includes built-in capabilities that go beyond detection and response.
    - Threat and Vulnerability Management (TVM) and
    - Attack Surface Reduction (ASR) work together to reduce risk proactively, before an attack occurs.

Threat and Vulnerability Management (TVM)

- Threat and Vulnerability Management (TVM) is a continuous, agent-based vulnerability assessment capability in MDE that **identifies, prioritizes, and helps remediate** software vulnerabilities and security misconfigurations across endpoints.
- Unlike traditional vulnerability scanners, TVM is continuous, real-time, and threat-aware.

Key Functions of TVM:

Key Functions of TVM:

- **Continuous Vulnerability Discovery**
  - Identifies:
    - OS vulnerabilities
    - Third-party application vulnerabilities
    - Missing patches
    - End-of-life software
  - No network scans required
- **Risk-Based Prioritization**
  - TVM prioritizes vulnerabilities based on:
    - Exploit availability
    - Active threat usage
    - Asset exposure
    - Business impact
  - This produces a Security Exposure Score.
- **Security Configuration Assessment**
  - Evaluates:
    - OS hardening settings
    - Defender configuration gaps
    - Credential protection status
    - Firewall and exploit protection settings
- **Remediation Recommendations**
  - Actionable guidance for:
    - Patching
    - Configuration changes
    - Application upgrades
  - Integrates with:
    - Microsoft Intune
    - Configuration Manager (SCCM)

Attack Surface Reduction (ASR)

- Attack Surface Reduction (ASR) is a set of preventive security controls designed to minimize the ways attackers can exploit endpoints by blocking high-risk behaviors commonly abused by malware.

- ASR focuses on **prevention**, not detection.

Core ASR Components

- **a. ASR Rules**
  - Behavior-based rules that block:
    - Office apps creating child processes
    - Credential theft from LSASS
    - Obfuscated PowerShell execution
    - Abuse of WMI and scripting engines
    - Executable content from email and web
  - **Modes**
    - Disabled
    - Audit
    - Block
    - Warn
- **b. Controlled Folder Access**
  - Protects critical folders from ransomware
  - Allows only trusted apps to modify protected data
- **c. Exploit Protection**
  - System-wide and app-specific exploit mitigations
  - Protects against memory corruption attacks
- **d. Network Protection**
  - Blocks outbound connections to malicious domains and IPs
  - Prevents command-and-control (C2) communication

TVM vs ASR

Aspect	TVM	ASR
Focus	Reduce vulnerabilities	Reduce attack paths
Nature	Visibility & prioritization	Prevention & blocking
Timing	Before attack	During attack
Output	Risk scores & recommendations	Blocked actions & alerts


That's all Folks!